

Agile Testing Documentation - DentWise Application

1. Project Overview

Project Name: DentWise - Dental Clinic Management System

Sprint Duration: 2 weeks

Team Size: 4 members

Methodology: Scrum

2. Sprint Backlog

Sprint 1: Core Testing Implementation

ID	User Story	Story Points	Priority	Status
US-01	As a QA engineer, I want to analyze requirements so that I can create comprehensive test cases	5	High	☒ Done
US-02	As a QA engineer, I want to design black-box test cases so that I can validate user-facing functionality	8	High	☒ Done
US-03	As a QA engineer, I want to perform white-box testing so that I can ensure code coverage	8	High	☒ Done
US-04	As a QA engineer, I want to write unit tests so that I can automate regression testing	5	High	☒ Done
US-05	As a QA engineer, I want to document Agile process so that the testing workflow is transparent	3	Medium	☒ Done
US-06	As a QA engineer, I want to report bugs so that developers can fix issues	3	High	☒ Done

Total Story Points: 32

3. User Stories Detail

US-01: Requirements Analysis

As a QA engineer

I want to analyze the DentWise application requirements

So that I can identify testable functional and non-functional requirements

Acceptance Criteria:

- ☐ Identify 5-7 functional requirements
- ☐ Identify at least 1 non-functional requirement
- ☐ All requirements are clear and testable
- ☐ Requirements are documented in markdown format

Story Points: 5

Priority: High

Sprint: 1

US-02: Black-Box Testing

As a QA engineer

I want to design black-box test cases using EP and BVA techniques

So that I can validate the application without knowledge of internal code

Acceptance Criteria:

- ☐ Design 8-10 test cases
- ☐ Apply Equivalence Partitioning technique
- ☐ Apply Boundary Value Analysis technique
- ☐ Document test cases with all required fields
- ☐ Execute tests and record results

Story Points: 8

Priority: High

Sprint: 1

US-03: White-Box Testing

As a QA engineer

I want to perform white-box testing on a selected function

So that I can ensure structural coverage of the code

Acceptance Criteria:

- ☐ Select one function for analysis
- ☐ Draw Control Flow Graph (CFG)
- ☐ Calculate Cyclomatic Complexity
- ☐ Identify independent paths
- ☐ Implement tests for path coverage

Story Points: 8

Priority: High

Sprint: 1

US-04: Unit Testing

As a QA engineer

I want to write automated unit tests using Jest

So that I can enable continuous regression testing

Acceptance Criteria:

- ☐ Write 6-10 unit tests
- ☐ Cover normal, edge, boundary, and invalid cases
- ☐ Achieve >90% code coverage
- ☐ All tests pass successfully

Story Points: 5

Priority: High

Sprint: 1

US-05: Agile Documentation

As a QA engineer

I want to document the Agile testing process

So that the testing workflow is transparent and reproducible

Acceptance Criteria:

- ☐ Create Sprint Backlog
- ☐ Write User Stories
- ☐ Define Testing Tasks
- ☐ Document Sprint Board status

Story Points: 3

Priority: Medium

Sprint: 1

US-06: Bug Reporting

As a QA engineer

I want to report bugs found during testing

So that developers can fix the issues

Acceptance Criteria:

- ☐ Report at least 2 bugs
- ☐ Include steps to reproduce
- ☐ Include expected vs actual results
- ☐ Assign severity and priority

Story Points: 3

Priority: High

Sprint: 1

4. Testing Tasks Breakdown

Task Board

Task ID	Task Name	Assigned To	Story	Status	Hours
T-01	Analyze application features	Team Lead	US-01	☐ Done	2
T-02	Document functional requirements	QA Engineer 1	US-01	☐ Done	3
T-03	Document non-functional requirements	QA Engineer 1	US-01	☐ Done	1
T-04	Create equivalence partitions	QA Engineer 2	US-02	☐ Done	2
T-05	Identify boundary values	QA Engineer 2	US-02	☐ Done	2
T-06	Design 10 test cases	QA Engineer 2	US-02	☐ Done	4
T-07	Execute black-box tests	QA Engineer 2	US-02	☐ Done	2
T-08	Select function for analysis	QA Engineer 3	US-03	☐ Done	1
T-09	Draw Control Flow Graph	QA Engineer 3	US-03	☐ Done	3
T-10	Calculate Cyclomatic Complexity	QA Engineer 3	US-03	☐ Done	1
T-11	Identify independent paths	QA Engineer 3	US-03	☐ Done	2
T-12	Write Jest unit tests	QA Engineer 4	US-04	☐ Done	4
T-13	Run coverage analysis	QA Engineer 4	US-04	☐ Done	1
T-14	Create Sprint documentation	Team Lead	US-05	☐ Done	2
T-15	Identify and log bugs	All Team	US-06	☐ Done	2

5. Sprint Board Visualization

SPRINT 1 BOARD: DentWise Testing

TO DO

IN PROGRESS

TESTING

DONE

US-01
Requirements
Analysis
(5 pts)

US-02
Black-Box
Testing
(8 pts)

US-03
White-Box
Testing
(8 pts)

US-04
Unit Testing
(5 pts)

US-05
Agile Docs
(3 pts)

US-06
Bug Reports
(3 pts)

Sprint Progress: 100%
Story Points: 32/32 completed

6. Sprint Metrics

Burndown Data

Day	Remaining Points	Ideal
1	32	32
2	28	29
3	24	26
4	20	23
5	16	19
6	12	16
7	8	13
8	5	10
9	3	6
10	0	3

Sprint Summary

Metric	Value
Total Story Points	32
Completed Points	32
Velocity	32 pts/sprint
Sprint Goal Achievement	100%
Test Cases Created	20
Test Cases Passed	20
Bugs Found	2
Code Coverage	97.29%

7. Daily Standup Template

Team Member: [Name]

Date: [DD/MM/YYYY]

Yesterday:

- [What was completed]

Today:

- [What will be worked on]

Blockers:

- [Any impediments]

8. Definition of Done (DoD)

For each User Story to be considered “Done”:

- ☐ All acceptance criteria are met
 - ☐ Code is written and committed
 - ☐ Unit tests are written and passing
 - ☐ Documentation is updated
 - ☐ Peer review is completed
 - ☐ No critical bugs remain
 - ☐ Product Owner has accepted the story
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9. Jira Configuration

Project Settings

- **Project Key:** DENT
- **Project Type:** Scrum
- **Board Name:** DentWise Testing Board

Issue Types

1. Epic
2. User Story
3. Task
4. Bug
5. Test

Workflow

To Do → In Progress → Testing → Done

Labels Used

- requirements
 - black-box
 - white-box
 - unit-test
 - bug
 - documentation
-

10. Sprint Retrospective

What Went Well

- All test cases were designed and executed successfully
- 100% branch coverage achieved
- Team collaboration was effective
- Documentation was comprehensive

What Could Be Improved

- Earlier integration with Jira/Xray
- More automated reporting
- Better time estimation for complex tasks

Action Items for Next Sprint

1. Set up automated test reporting
2. Create test data management strategy
3. Implement CI/CD pipeline integration